

A Rate and a Starting Value



Grade 8 • Mathematics
NC.8.F.4

Build and interpret linear models.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A club's total cost y dollars depends linearly on x kits. Two records are $(5,47)$ and $(8,71)$. Find $y = mx + b$ and interpret m and b .

$m = (71 - 47) / (8 - 5) = 8$ dollars/kit. $b = 47 - 8(5) = 7$ dollars before any kits.
Plot $(0,7)$ and the two records.



Build a second model together

A club's total cost y dollars depends linearly on x kits. Two records are $(3,8)$ and $(6,14)$. Find $y = mx + b$ and interpret m and b .

Discuss: What should the first step find?



Your turn • 1

A club's total cost y dollars depends linearly on x kits. Two records are $(3,19)$ and $(6,34)$. Find $y = mx + b$ and interpret m and b .

Represent → solve → explain → check



Your turn • 2

A club's total cost y dollars depends linearly on x kits. Two records are $(5,16)$ and $(8,22)$. Find $y = mx + b$ and interpret m and b .

Represent → solve → explain → check



Your turn • 3

A club's total cost y dollars depends linearly on x kits. Two records are $(7,24)$ and $(10,33)$. Find $y = mx + b$ and interpret m and b .

Represent → solve → explain → check



Your turn • 4

A club's total cost y dollars depends linearly on x kits. Two records are $(5,14)$ and $(8,20)$. Find $y = mx + b$ and interpret m and b .

Represent → solve → explain → check



Exit question • Explain independently

A club's total cost y dollars depends linearly on x kits. Two records are $(5,22)$ and $(8,31)$. Find $y = mx + b$ and interpret m and b .

Include a check. What would you teach someone else?



Teacher review and next steps

1. $y = 5x + 4$

2. $y = 2x + 6$

3. $y = 3x + 3$

4. $y = 2x + 4$

Exit: $y = 3x + 7$

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

